



## Make Your Own ESP8266-12E/F Module Programmer

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**T**his article describes a method for making a programmer for ESP8266-12 E/F Wi-Fi chip and also discusses the support circuitry required for real-time applications.

NodeMCU board is a development kit that consists of ESP8266 Wi-Fi-enabled chip and many other support circuitries. It is a popular board, which is handy to work with Wi-Fi-related projects. But when product development and testing stages are completed, the whole NodeMCU board is not required—only ESP8266 chip is required for the final product. This reduces the overall cost of the product, lowers power consumption, and ensures a smaller PCB footprint and code security.

However, it is not easy to separate ESP8266 chip from NodeMCU board. An alternate solution is to buy ESP8266 chip separately, which is also much cheaper than NodeMCU board. With this option, you can directly burn the code into ESP8266 chip using a suitable programmer, like the one described in this article. After that, you can pull out ESP8266 chip from the programmer and use it in your final product.

The programmer used by the author for ESP8266-12 E/F module is shown in Fig. 1.

The process of designing and building a project using ESP8266 can be broken down into two distinct steps: hardware and software. Hardware revolves around building support circuitry to GPIOs, while software revolves around writing, compiling and loading the sketch into the chip. Such application again involves two stages: development and deployment.

### Development stage

During development stage, you need a software development kit (SDK),

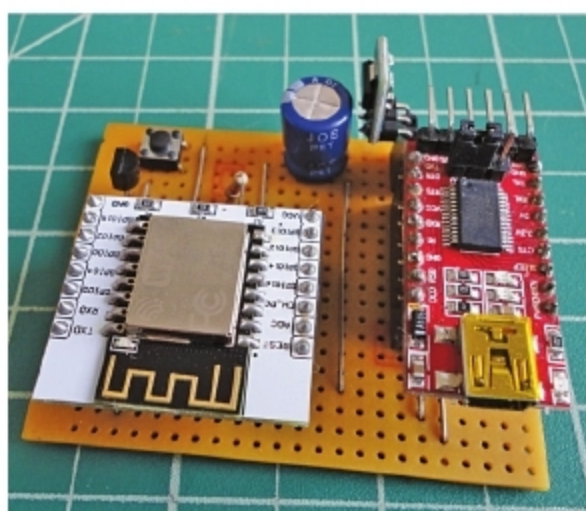


Fig. 1: Programmer used by author

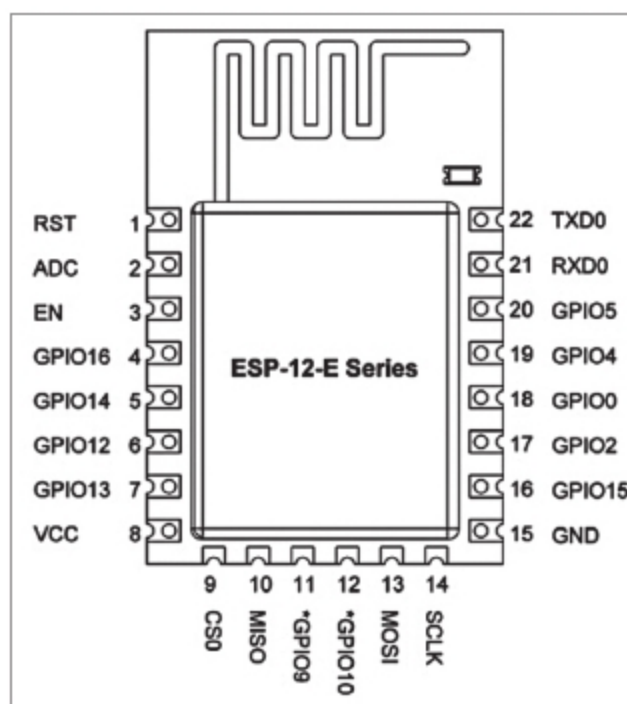


Fig. 2: Pin details of ESP-12E module



Fig. 3: Front and rear sides of breakout board for ESP-12E/F chip